

B.Tech. Degree IV Semester Regular/Supplementary Examination June 2023

ME 19-205-0403 MECHATRONICS
(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

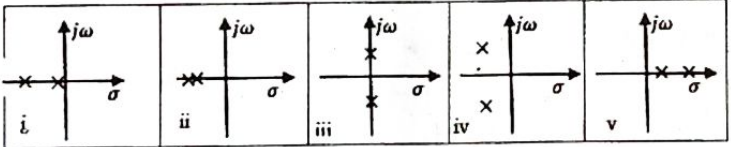
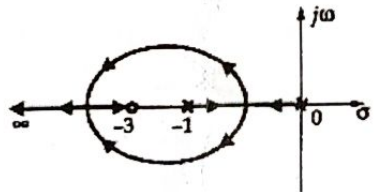
Course Outcomes

On successful completion of the course, the students will be able to:

- CO1: Understand the basics of sensors and transducers and different mechatronics systems.
 CO2: Model first and second order systems using block diagrams and governing equations.
 CO3: Analyse first and second order systems for their time and frequency domain responses and stability.
 CO4: Design mechatronics system for simple application.
 CO5: Use software tools for modelling and simulation of mechatronic systems.
 Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create
 PO – Programme Outcome

PART A

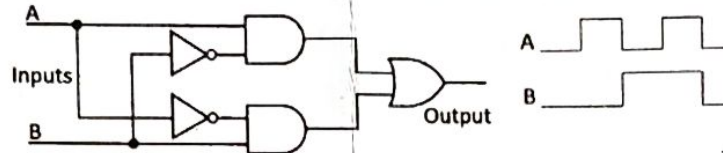
(Answer ALL questions)

- (8 × 3 = 24) Marks BL CO PO
1. (a) Describe the working of a pneumatic proximity sensor. 3 L1 1 1
 (b) Draw the schematic representations of different actuation methods in direction control valves. 3 L2 1 2
 (c) A unity negative feedback system has an open loop transfer function $G(s) = \frac{k}{s(s+1)}$. What should be the value of gain K for the system to have a damping ratio of 0.025? 3 L3 3 3
 (d) The pole locations of a second order system are shown below. Sketch the response of the system to a unit step input in each case. 3 L3 3 3
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- (e) The root locus of a unity negative feedback system is shown below. What will be open loop and closed loop transfer functions. Justify your answer. 3 L4 3 3
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- (f) The open loop transfer function of a unity negative feed back system is $G(s) = \frac{k}{s(s+1)(s+3)}$. Find the value of K (greater than 0) for which the system will be marginally stable. 3 L4 3 4

(P.T.O.)

Marks	BL	CO	PO
3	L2	1	1
3	L5	3	2

- (g) Describe how a signal is latched in ladder program.
 (h) For the logic gate system given in figure, draw the output.



PART B

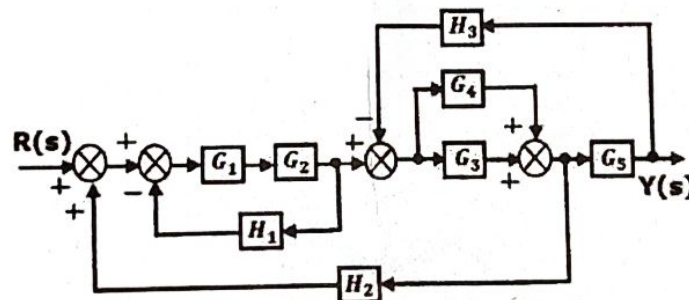
(4 × 12 = 48)

- II. (a) Explain the working of
 (i) absolute encoder
 (ii) hall effect sensor
 (iii) gear pump
 (b) Two double-acting pneumatic cylinders A and B are to be sequenced as follows:
 A and B extends together; then B retracts; then A retracts.
 Draw a layout for this using suitable DCVs and limit switches and explain the working.
 (c) Describe the function of pressure relief valve and represent it schematically.

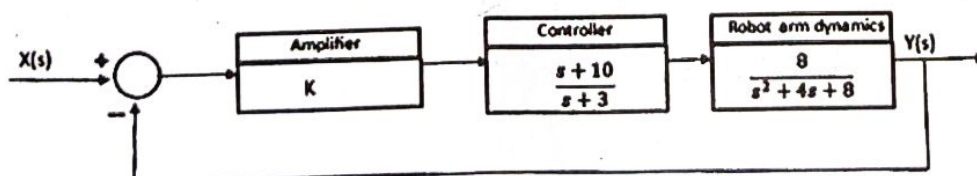
OR

- III. (a) Describe the working and use of:
 (i) LVDT.
 (ii) Triac.
 (iii) bonded strain gauge.
 (b) Explain the working of permanent magnet and variable reluctance stepper motors.
 (c) A d.c. motor is connected to a 24 V d.c. source. The voltage applied across the motor has to be brought down to 6 V to control its speed. Represent the PWM signal and what would be the duty cycle.

- IV. (a) Find the overall transfer function of the given system shown.



- (b) Robots are used in manufacturing and assembly operations that require accurate, fast, and versatile manipulation. A typical robotic arm system is shown in figure. Determine the value of K for which the system will be stable.

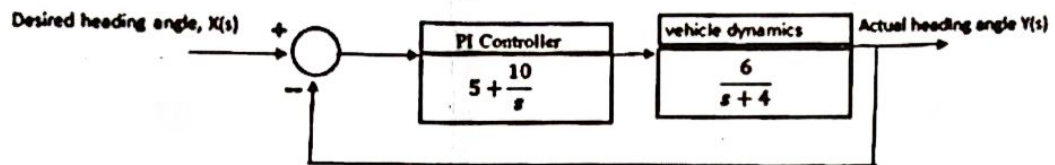


OR

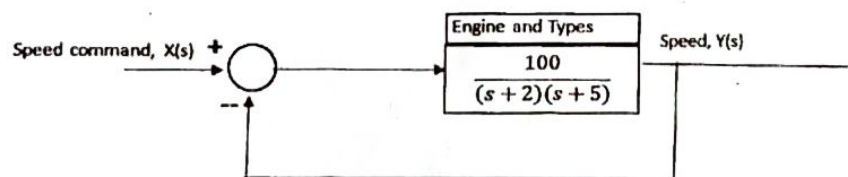
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	Marks	BL	CO	PO
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- V. (a) A severely disabled person uses a mobile robot as an assisting device, the steering control of which is represented by the block diagram shown. Determine the steady state error of the system for
- a step input of $x(t) = 1.5$ radian
 - a ramp input of $x(t) = 20t$ radian



- (b) The engine, body, and tyres of a racing car affect the acceleration and the speed attainable. The speed control of the car is represented by the model shown in figure. Calculate the following.
- Rise time.
 - Peak time.
 - percentage overshoot.
 - 2% settling time.



- VI. In a space station, it is critical to keep the station in proper orientation toward the sun and earth for generating power and communications. The orientation controller may be represented by a unity negative feedback system with an actuator and controller with $G(s) = \frac{k(s+20)}{s(s+12)^2}$
- Plot the root locus.
 - What is your inference regarding the stability of the system from this plot?
 - What is the value of K for which the response becomes oscillatory (Complex poles)?

OR

- VII. The open loop transfer function of a robot arm joint-control is given as $G(s) = \frac{300(s+100)}{s(s+10)(s+40)}$
- Plot the Bode diagram (Magnitude and Phase).
 - Determine the gain cross-over frequency, phase cross over frequency, gain margin and phase margin.
 - What is the inference that you can draw from the gain margin?

(P.T.O.)

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		Marks	BL	CO	PO
VIII.	(a) The entry to a theatre is restricted using a PLC controlled automatic door. When a token is inserted into a slot, the door opens. A sensor placed at the doorway detects the passage of persons. After three persons pass through the door, it closes. Suggest a suitable actuation method for the door, draw the layout diagram and write the PLC program.	6	L5	4	4
	(b) Draw the internal architecture of a microprocessor and explain the various registers and components in it.	6	L2	1	1
OR					
IX.	(a) A scheme is devised to sort coins on the basis of colour and shape. All bronze-coloured coins are to be placed in Bin-X. Silver coloured coins that have round shape should go to Bin-Y and that have square shape should go to Bin-Z. If the above combinations are exhaustive, represent the scheme using logic gates, clearly mentioning the states of inputs and outputs.	6	L5	4	4
	(b) A pump, when switched on, pumps water to a container until a sensor gives a signal that the required level has reached. Then a heater heats the water for 10 minutes. Write a PLC program for this.	6	L5	4	2

Bloom's Taxonomy Levels

L1 – 12.5%, L2 – 10%, L3 – 14.17%, L4 – 22.50%, L5- 40.83%.
